Interactive Dashboard

Build a population simulator (Step 1)

Textbooks frequently contain static plots/images demonstrating concepts. For some students, dynamic interaction with the material can be critically helpful in internalizing those concepts. To demonstrate one example, do the following:

1) Copy and paste the following text into your favorite LLM.

   I teach high-school ecology. Build me a single, self-contained HTML page that simulates a deer population over 50 years, for students to use in class.

   What it must do:

   1. Sliders for three deer settings: growth rate per year, carrying capacity (K), and starting population. Show the current value next to each slider.

   2. One graph with years on the horizontal axis and deer on the vertical axis, showing two curves at once: exponential growth (no limit) as a dashed line, and logistic growth (limited by K) as a solid line. Draw the carrying capacity as a labeled dashed horizontal line.

   3. A checkbox labeled "Add wolves." When it is on, add a wolf population that eats deer (predator–prey dynamics), with sliders for starting wolves, hunting success, and wolf death rate. Plot wolves on the same graph using a second axis on the right so both populations are readable.

   4. Under the graph, a short readout: deer at year 50 under each model, the year the herd first reaches 90% of K, and two or three plain sentences explaining what the current settings show.

   5. Everything redraws instantly as sliders move. Include a "Reset to defaults" button.

   Choose default values so that, with wolves on, the two populations visibly cycle. Keep the

page clean and readable on a classroom projector, with large labels. Use no external libraries; draw the graph yourself with SVG. At the end, tell me briefly which equations you used.

2) Download the resulting HTML file (assuming it makes one properly), and open it (you can just "double click" and it should open in a web browser).
3) Play with the tool it developed.
4) Ask the LLM for one modification (a change in color scheme; more detailed explanations; etc.)

Reflection points:

- Did it work?
- Would a tool like this be useful in your instructional settings?
- Would you write a prompt like this yourself?

Build a population simulator prompt (Step 2)

First, reflect upon the following question: Could/would you have written the prompt above? My answer: "Likely no". I understand the prompt – and it seems reasonable to me – but I would not have written this prompt myself.

Now for the good news: I didn't write that prompt myself either. Instead, *I used an LLM to build/refine this prompt*, and you can also. Let's try it out.

Instead, now, do these four things.

1) Download the Biology II: Ecology VDOE Instruction Plans here ideally as a PDF.
2) Upload that document (again, ideally, a PDF) into a fresh LLM chat window.
3) Add this prompt: "I've attached an instructional guide to this chat. Look at standard EC.5, about population growth, carrying capacity, and limiting factors. Students find this hard because the textbook shows one finished graph, and they can't see what happens when the growth rate, carrying capacity, or a predator changes. Write me a prompt I can paste into a new chat that will produce a simple interactive web page (i.e. an HTML file) that students can use to explore this. The page should have sliders and a graph, need no files or programming on my part, and work on the first try. Please try to keep the prompt brief; I don't want to be overwhelmed checking it."
4) Try it out.

Reflection points:

- Did it work?
- Would you write a prompt like this?

- How might you adopt this to your own teaching domain?

Key Takeaways

1) Prompt design can start simple. Just point

Interpreting a Messy Plot

Clean up a messy plot – Hemophilia

Not every confusing figure is a plot. The pedigree of hemophilia among Queen Victoria's descendants is the standard textbook example of X-linked inheritance, and it is nearly unreadable: four generations, five royal houses, sixty-plus symbols, and marriages that cross between branches. The rule students need is simple; the diagram is cool because it shows a complex system in which the rule dominates, but the complexity of the system hides the simplicity of the rule.

1. Go to the Wikipedia article "Haemophilia in European royalty" and save the family-tree image (right-click, save image). Any textbook version of the same pedigree works too.
2. Upload the image to a fresh LLM chat window.
3. Add this prompt:

   I've attached the pedigree of hemophilia in Queen Victoria's descendants. My high-school students can't see the inheritance rule in it. Pick one branch that shows X-linked recessive inheritance most clearly: a carrier mother, at least one affected son, and at least one unaffected son, across three generations. Redraw just that branch as a clean pedigree with standard symbols (squares for males, circles for females, filled for affected, half-filled for carriers), write the genotype under each person using X^H and X^h, and add one sentence under the diagram stating the rule the branch demonstrates. Tell me which people you chose and why. Critically: do not "generate" an image, but instead use code to create your result so that it is reproducible and modifiable.

Reflection points:

- Can you verify this new visualization is legitimate based on the original figure? Or, what steps *must* you take before using such a modified visualization in an educational context?
- Is there a context where such a simplified plot would be useful on a text?

Clean up a messy plot – CO2 data over time

The NOAA Mauna Loa CO_2 record (the "Keeling curve") is the most famous graph in environmental science. It is also confusing to students: a 65-year rise with a jagged yearly wiggle on top, and they can't tell which part matters. In fact both parts matter, for different reasons, and that is the problem.

Take the following steps:

1) We're going to use real data here – but you don't need to download it. Instead, simply note that the data is from this NOAA official government website (https://gml.noaa.gov/webdata/ccgg/trends/co2/co2_mm_mlo.csv), and pay attention to how I reference it below.
2) Upload it to a fresh LLM chat window.
3) Add this prompt:

   The data from NOAA's monthly CO_2 data from Mauna Loa can be found here: https://gml.noaa.gov/webdata/ccgg/trends/co2/co2_mm_mlo.csv

   The usual plot of this file confuses my high-school students because two different patterns are mixed together. Make me three clean, separate charts: (1) the normal plot that shows the data as a function of time, (2) the normal plot, but now showing data as a function of time with the month of the year shown with a smoothly varying color scheme where winter months are more blue, and hot months are more red, (3) the seasonal cycle, month by month, for all years with labels saying what plants in the Northern Hemisphere are doing in spring, summer, and fall, and (4) just the long-term rise since 1958, with the yearly wiggle removed. Tell me which columns of the file you used.

Reflection points:

- Can you trust these new plots? Or, what steps *must* you take before using such a modified visualization in an educational context?
- Would a tool like this be useful?
- Would you write a prompt like this?

Plot/Graph Interpretation

Interpretation practice (round 1)

The previous exercises made visualizations. This one makes the questions. The standards require students to construct and interpret data tables and graphs, and writing good practice data with a clean answer key takes real time. This is a job an LLM does well.

4. Open a fresh LLM chat window (no files needed).
5. Add this prompt:

   I teach high-school genetics. Make me a one-page worksheet on chi-square analysis of a genetic cross.

   Use a dihybrid corn cross (purple/yellow kernel color, smooth/wrinkled texture) and invent a realistic set of observed counts for about 400 kernels that is close to a 9:3:3:1 ratio but not exactly. Give students: (1) the observed counts in a table with the expected column blank, (2) a bar chart comparing observed and expected for the four phenotypes, and (3) six questions that build from easy to hard: read a value from the chart, calculate expected counts, calculate chi-square, find degrees of freedom, compare to the critical value at p = 0.05, and finally write one sentence stating whether the data support independent assortment and what it would mean if they did not.

   Then give me a separate answer key with the worked arithmetic and a short list of the three mistakes students most often make on this topic. Keep the student page clean enough to print.

   Ensure any documents you give me are editable in Microsoft Word format.

6. Try it out.
7. Ask a follow-up: "Now make a second version of the data where the chi-square test fails, so students can see what a bad fit looks like."
8. Ask another follow up: "Now, make another version of this with deeper annotations of what is required at each step so that students struggling with this concept have a better chance of successfully working through the problem with guidance."
9. Ask another follow up: "Now, make another version with increased difficulty so that I can challenge students who found the original version straightforward."

Reflection points:

- Is the arithmetic in the answer key right? (Check some numbers yourself.)
- Would you hand the student page out as is, or edit it first? What would you change?

Interpretation practice (round 2)

4) Upload the Biology II: Genetics VDOE instructional guide (PDF) into a fresh LLM chat window.
5) Add this prompt:

   I've attached an instructional guide. Look at standard GEN.5 a. Students struggle to understand what "fit" means, even when they can do the arithmetic. Write me a short prompt I can paste into a new chat that will produce a one-page practice worksheet with realistic data, a chart, questions that build from easy to hard, and a separate answer key. Keep the prompt brief.

6) Try it out.
7) Compare the prompt it wrote to the one in Step 1. Compare the results.
8) Then swap "GEN.5 a" for a standard you actually teach and run it again.

Reflection points:

- Did it work?
- Which standard from your own course would you try this on next?
- What would you need to check before trusting an AI-written answer key with your students? Relatedly: Does the use of AI offload any instructor responsibility for the content of the worksheet and answer key, should it be used in an educational setting?